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Part 1

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Task 1: Research

Introduction

Internet of Things (IoT) dashboards are used to show the data that is collected from different connected devices and sensors. When there are many sensors sending information at the same time, it can be difficult for a user to notice important changes or problems quickly. Because of this, an IoT dashboard should not only display the data, but should also make the information clear and easy to understand. This research looks at five different user engagement strategies that can be used to improve an IoT dashboard and make it easier for users to identify problems and interact with the system.

5 User Engagement Strategies for IoT Dashboards

1. Real-Time Data Visualization:

Real-time data visualisation allows users to see sensor information as it changes instead of waiting for the data to be updated later. This is useful in IoT systems because sensors can send new readings very often. Showing the latest values on charts, graphs or status cards can help users quickly understand what is happening in the system. It can also make it easier to notice sudden changes, such as a temperature increasing too quickly or a device producing unusual readings. For a Smart-X dashboard, real-time visualisation would help users monitor different sensors from one place and react faster when something changes.

2. Anomaly and Alert Notifications

Anomaly and alert notifications are used to draw attention to readings or events that may need immediate action. In an IoT system, this can include things such as a sensor going offline, a temperature reading becoming too high or a device sending unusual values. Instead of expecting the user to constantly watch every sensor, the dashboard can highlight important problems and show a warning when something is outside the normal range. This can help users respond faster and makes it easier to manage a large number of connected devices. In the Smart-X system, alerts could be used to show when a sensor has a critical reading or when a device has stopped sending data.

3. Interactive Filtering and Drill-Down

Interactive filtering and drill-down features allow users to focus on the information that is most relevant to them. In an IoT dashboard, there may be many devices and sensor readings shown at the same time, which can make the screen difficult to understand. Filters can allow users to view information based on things such as device type, location or status. Drill-down features can then give more detailed information about a specific sensor when the user selects it. This can make the

dashboard easier to use because users do not have to look through all the data at once. In the Smart-X system, users could filter sensors by category or location and then open a selected sensor to view its latest readings and details.

4. Device Health and Status Indicators

Device health and status indicators are used to show whether a connected device is working correctly or if there may be a problem. These indicators can use simple labels such as online, offline, warning or critical, which makes it easier for users to understand the current state of a sensor without checking all of its readings. This is useful in IoT systems because a device may stop responding even if there is no obvious problem shown in the latest data. In the Smart-X system, status indicators could help users quickly identify sensors that are disconnected, inactive or producing unusual readings.

5. Contextual Help and Troubleshooting Guidance

Contextual help and troubleshooting guidance can make an IoT dashboard easier to use, especially when users are not sure what a warning or sensor reading means. Instead of only showing that there is a problem, the dashboard can also give a short explanation or suggest what the user should check next. For example, if a sensor is offline, the system could suggest checking its connection or power source. This can help users solve smaller problems faster without needing to search for instructions somewhere else. In the Smart-X system, this type of guidance could be shown next to warnings or device status messages.

Selected Strategy: Anomaly and Alert Notifications

The strategy that I would choose for the Smart-X dashboard is anomaly and alert notifications. This strategy is useful because IoT systems can collect a large amount of sensor data, which can make it difficult for users to notice important problems quickly. Alerts can help bring attention to unusual readings, disconnected devices or values that fall outside a normal range. This makes the dashboard more useful because users do not have to constantly monitor every sensor manually.

Anomaly and alert notifications can improve user engagement because they make the dashboard feel more responsive and useful. Instead of the user only seeing large amounts of sensor data, the system can point out which information needs attention. This helps reduce the amount of time the user has to spend checking every reading manually. It can also make the system easier to understand because warnings can clearly show which devices are operating normally and which ones may need to be checked. This can encourage users to interact with the dashboard more often because the information is easier to follow and more relevant.

In the Smart-X system, anomaly and alert notifications could be linked to the readings received from each registered sensor. The system could compare the latest reading to a normal range that has been set for that sensor type. If the value moves

outside of this range, the dashboard could display a warning or critical alert. For example, a temperature sensor could show a warning if the reading becomes too high, while a device that stops sending data could be marked as offline. These alerts could appear on the main dashboard so that the user can immediately see which devices need attention.

One of the main benefits of using anomaly and alert notifications is that they can help users respond to problems faster. They can also reduce the need to constantly check every sensor manually. However, if too many alerts are shown, users may start ignoring them, especially if some warnings are not important. Because of this, the Smart-X system should make sure alerts are only shown when they are useful and should clearly separate normal warnings from more serious problems. This can help prevent the dashboard from becoming confusing or overwhelming.

The design of the alerts is also important because users need to understand the seriousness of a problem quickly. The Smart-X dashboard could use clear labels such as normal, warning and critical, together with different visual indicators so that users can tell the difference between them. Alerts should also include a short explanation of what happened and which sensor is affected. This can make the dashboard easier to use because the user does not have to search through multiple pages to understand the problem.

Overall, anomaly and alert notifications would be a suitable engagement strategy for the Smart-X dashboard because they help users focus on the most important information. The strategy can make large amounts of sensor data easier to manage by highlighting unusual readings and device problems as they happen. If the alerts are designed clearly and only show useful information, they can improve the usability of the dashboard and help users react to issues more quickly.

Conclusion

User engagement is important when designing an IoT dashboard because users need to understand sensor information quickly and clearly. Strategies such as real-time visualisation, alerts, filtering, device status indicators and troubleshooting guidance can make a dashboard easier to use. For the Smart-X system, anomaly and alert notifications would be especially useful because they can help users identify unusual readings and device problems without having to constantly monitor every sensor manually.

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